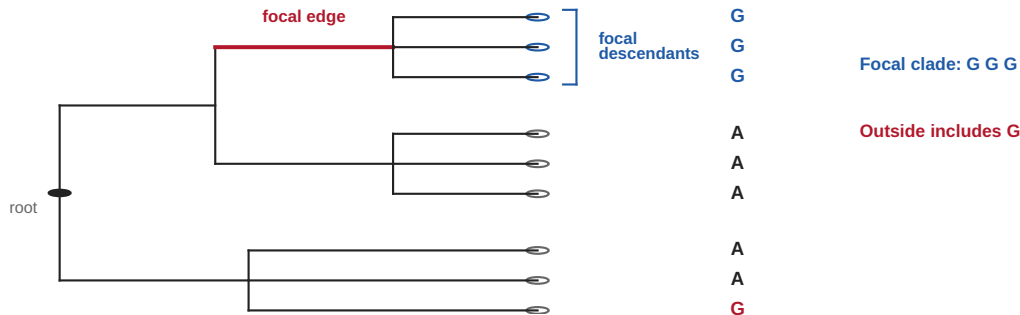


A One branch, one site



B Two questions, different answers

OBSERVED

Clade-exclusive?

focal G G G

outside A A A A A G

NO not fixed-exclusive
G occurs outside the clade

≠

kept separate

INFERRED

Substitution on the focal edge?

all global optima

parent child

A → G

YES unambiguous A → G
 $P^* = \{A \rightarrow G\}$

C Retain all optimal focal-edge solutions

Optimal pair set

BRANCHSNV reports

$P^* = \{A \rightarrow G\}$ Unambiguous change

$P^* = \{A \rightarrow G, C \rightarrow G\}$ State ambiguity

$P^* = \{A \rightarrow G, G \rightarrow G\}$ Placement ambiguity

$P^* = \{G \rightarrow G\}$ No change

No arbitrary tie-breaking

All globally optimal focal-edge state pairs are retained.

DESIGN SAFEGUARDS

Explicit root
defines direction

Exact descendants
defines branch

Exact taxon labels
defines correspondence

Complete optimal set
retains uncertainty

Deterministic provenance
supports auditability